

1 **Supplementary Appendix: Apparent Catamenial Epilepsy Classification Under Independence**

2 This appendix provides additional methodological details, defined analysis choices, quality-control checks, sensitivity analyses, and
3 supplementary results for the null simulation. It is intended as online supplementary material accompanying the main Neurology
4 submission draft.

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14 **A1. Simulation design and reproducibility**

15 The analysis used a deterministic master seed of 20260505. The full run simulated 100,000 synthetic participants, processed in
16 chunks, with 50,000 in the healthy ovulatory cohort and 50,000 in the heterogeneous menstruating-age cohort. The 36-month diaries

17 were generated independently for seizures and menstrual/hormone cycles, followed by a participant-specific random circular shift
18 before merging.

19 Daily audit data were saved for a 1% random sample. Participant-level and window-level outputs were written as parquet and CSV
20 files, and every output artifact was recorded in a machine-readable manifest with file size and SHA-256 checksum.

21 The term 'defined before the full simulation run' is used instead of 'prespecified' unless a dated protocol, locked commit, or
22 preregistration is supplied. The manifest records the seed, config values, output checksums, and adapter assumptions.

23 **A2. Simulator descriptions and validation targets**

24 CHOCOLATES generated seizure-count diaries independently of menstrual-cycle state. The adapter records observed seizure burden
25 and the simulator-reported dominant seizure-cycle period when available. Because multidien rhythms near monthly periods are a key
26 potential mechanism for chance alignment, submitted versions should report the distribution of dominant_seizure_cycle_days and
27 sensitivity analyses excluding or flattening 24-35-day seizure rhythms.

28 HORMONE-CYCLE generated menstrual/hormone diaries independently of seizure state. The adapter records age, cycle length,
29 within-participant cycle-length SD, ovulatory fraction, progesterone and estradiol when available, and inadequate-luteal-phase flags
30 based on a progesterone threshold of 5.0 ng/mL. Submitted versions should include package versions or commits for both simulators
31 and a validation table comparing simulated diary metrics with empirical benchmarks.

32 Supplementary Table 1. Minimum simulator validation targets for submission.

Domain	Simulated metric to report	Reviewer-facing purpose
Seizure burden	Seizure days/month, seizures/month, interseizure interval dispersion, seizure-cluster metrics	Shows the null seizure diary is not an unrealistically sparse or overclustered artifact.
Seizure rhythms	Dominant multidienn period distribution; proportion in 24-35-day band; sensitivity excluding that band	Separates finite-count ratio artifacts from monthly-ish seizure-rhythm alignment.
Menstrual cycles	Cycle-length distribution, within-participant cycle-length SD, luteal length, ovulatory fraction, anovulation frequency	Shows phase labeling and C3 eligibility are generated from plausible cycle structure.
Missing/irregular behavior	Unlabeled days, non-23-35-day cycles, ILP/anovulatory flags, medical-factor prevalence	Quantifies sources of indeterminate windows and C3 sensitivity.

A3. Cohort definitions and diary generation

The healthy ovulatory cohort targeted adult ovulatory cycling, age 18-45 years, with simulator medical modifiers disabled where exposed. The heterogeneous menstruating-age cohort targeted a broader menstruating-age range of 13.0-54.9 years and allowed anovulation and irregularity. Because the hormone simulator exposed medical-factor controls but not a natural prevalence sampler, PCOS, peri-menarche, perimenopause, and dysmenorrhea were sampled from config.yaml rates. This cohort is an assumption-driven heterogeneity stress test unless external prevalence calibration is added.

Supplementary Table 2. Cohort construction parameters.

Parameter	Healthy ovulatory	Heterogeneous menstruating-age
Target sample size	50,000	50,000
Age range	18.0-45.0 years	13.0-54.9 years
Ovulatory-cycle handling	Ovulation probability set to 1.0	Natural ovulatory/anovulatory behavior allowed
PCOS rate	Disabled	10%

Parameter	Healthy ovulatory	Heterogeneous menstruating-age
Peri-menarche rate	Disabled	60% when age <20 years
Perimenopause rate	Disabled	55% when age >=45 years
Dysmenorrhea rate	Disabled	12%

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42 **A4. Phase labeling and catamenial epilepsy definitions**

43 Herzog-style phases were assigned on the full diary before window subsetting. For a complete cycle of length L, day 1 was the onset
 44 of menstrual flow and backward day was defined as d-(L+1). Phase assignment used an ordered lookup: menstrual phase if days 1-3 or
 45 backward days -3 to -1; follicular phase if days 4-9; ovulatory phase if day >=10 and backward day <=-13; luteal phase if backward
 46 days -12 to -4. The ovulatory condition is the intersection of forward and backward criteria. Days not satisfying any rule were
 47 unlabeled.

48 Supplementary Table 3. Implemented catamenial epilepsy definitions.

Definition	Primary rule	Main analyzability condition
Exact Herzog 2004	Three-cycle legacy rule; any CE if at least 2 of 3 cycles show any C1, C2, or C3 pattern.	Only complete 3-cycle windows with strict 23-35 day cycles.
Windowed Herzog thresholds	C1 $ADSF(M)/ADSF(F+L) \geq 1.69$; C2 $ADSF(O)/ADSF(F+L) \geq 1.83$; C3 $ADSF(O+L+M)/ADSF(F) \geq 1.62$ among ILP-flagged days.	Required phases observed; undefined ratios marked indeterminate.
Minimum-data rule	Windowed Herzog thresholds after applying minimum duration and seizure-burden criteria.	At least 4 months or 6 complete cycles, plus at least 4 seizure days.
Cycle reproducibility	At least two-thirds of eligible cycles positive for the same pattern and pooled ratio passes threshold.	At least 6 complete cycles for primary rule; 12 cycles for sensitivity.

Definition	Primary rule	Main analyzability condition
Negative-binomial regression	Daily GLM with M and O indicators; one-sided Holm-adjusted tests and C1/C2 rate-ratio thresholds. Stabilized and window-only dispersion versions reported.	Minimum-data rule passed; Poisson robust fallback recorded when needed.
Historical rules	Assumption-based operationalizations of Newmark-Perry, Duncan 1993, Herzog 1997 twofold, and Reddy 2007 any-phase rules.	Reported separately and flagged as assumption based.

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50 C3 was evaluated only in the heterogeneous cohort when `ilp_flag` was true. Anovulatory cycles without ILP labeling were not
51 automatically considered C3-positive or C3-applicable. Supplementary sensitivity rows report any CE excluding C3 and C1, C2, and
52 C3 separately.

53 **A5. Observation windows and analyzability rules**

54 Calendar windows used one random valid start day per participant for 1, 3, 4, 6, and 12 months. Cycle windows used one random
55 valid start among complete cycles for 3, 6, and 12 cycles. The full-window analysis used the entire 36-month diary. Windows were
56 marked indeterminate, not negative, when required phases or denominator information were unavailable.

57 Exact Herzog 2004 is rule-defined only for complete 3-cycle windows with strict 23-35-day cycles. Other window settings are not
58 evaluated for this rule; they should not be interpreted as negative evidence.

59 **A6. Statistical analysis and edge-case handling**

60 All ratio definitions used average daily seizure frequency rather than raw counts. If a comparator phase had observed days but zero
61 seizures, a positive numerator produced an infinite ratio and counted positive; zero numerator and zero comparator produced an

undefined ratio and an indeterminate result. Wilson 95% confidence intervals were used for simple proportions. Study-level Monte Carlo sampled 10,000 studies of 30 participants per cohort for the 3-month window.

Negative-binomial regression used daily rows with an intercept, menstrual indicator, ovulatory indicator, and cycle fixed effects when at least four complete cycles were available. The offset was one observed day. The Holm family comprised the menstrual and ovulatory one-sided tests within participant-window. Positivity required adjusted $P < 0.05$ and the corresponding threshold ratio. The stabilized-dispersion analysis uses full-diary alpha and should be interpreted as an oracle/stabilized comparator for short windows; the window-only alpha analysis is a non-oracle sensitivity.

Monte Carlo binomial intervals quantify uncertainty under the fixed simulation model. They do not include uncertainty about simulator validity, medical-factor prevalence, seizure-rhythm periods, or menstrual-cycle parameterization.

A7. Supplementary results

Supplementary Table 3. Window-length sensitivity for core definitions.

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	1 month	Cycle reproducibility rule	0	0	NA	100.0%
healthy ovulatory	1 month	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	1 month	Negative-binomial regression, stabilized dispersion	0	0	NA	100.0%
healthy ovulatory	1 month	Windowed Herzog thresholds	38,234	18,249	47.7%	23.5%
healthy ovulatory	1 month	Windowed Herzog with minimum data	0	0	NA	100.0%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	3 months	Cycle reproducibility rule	0	0	NA	100.0%
healthy ovulatory	3 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	3 months	Negative-binomial regression, stabilized dispersion	0	0	NA	100.0%
healthy ovulatory	3 months	Windowed Herzog thresholds	45,280	18,696	41.3%	9.4%
healthy ovulatory	3 months	Windowed Herzog with minimum data	0	0	NA	100.0%
healthy ovulatory	4 months	Cycle reproducibility rule	0	0	NA	100.0%
healthy ovulatory	4 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	4 months	Negative-binomial regression, stabilized dispersion	37,641	1,236	3.3%	24.7%
healthy ovulatory	4 months	Windowed Herzog thresholds	46,381	17,894	38.6%	7.2%
healthy ovulatory	4 months	Windowed Herzog with minimum data	37,641	13,501	35.9%	24.7%
healthy ovulatory	6 months	Cycle reproducibility rule	6,706	704	10.5%	86.6%
healthy ovulatory	6 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	6 months	Negative-binomial regression, stabilized dispersion	41,136	1,881	4.6%	17.7%
healthy ovulatory	6 months	Windowed Herzog thresholds	47,489	16,185	34.1%	5.0%
healthy ovulatory	6 months	Windowed Herzog with minimum data	41,136	12,985	31.6%	17.7%
healthy ovulatory	12 months	Cycle reproducibility rule	15,964	246	1.5%	68.1%
healthy ovulatory	12 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	12 months	Negative-binomial regression, stabilized dispersion	45,266	2,207	4.9%	9.5%
healthy ovulatory	12 months	Windowed Herzog thresholds	48,775	12,419	25.5%	2.4%
healthy ovulatory	12 months	Windowed Herzog with minimum data	45,266	10,661	23.6%	9.5%
healthy ovulatory	3 cycles	Cycle reproducibility rule	0	0	NA	100.0%
healthy ovulatory	3 cycles	Exact Herzog 2004, any pattern	23,101	11,616	50.3%	53.8%
healthy ovulatory	3 cycles	Negative-binomial regression, stabilized dispersion	17	Suppressed	Not interpreted (<1,000)	100.0%
healthy ovulatory	3 cycles	Windowed Herzog thresholds	45,122	18,831	41.7%	9.8%
healthy ovulatory	3 cycles	Windowed Herzog with minimum data	17	Suppressed	Not interpreted (<1,000)	100.0%
healthy ovulatory	6 cycles	Cycle reproducibility rule	22,591	2,742	12.1%	54.8%
healthy ovulatory	6 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	6 cycles	Negative-binomial regression, stabilized dispersion	40,861	1,860	4.6%	18.3%
healthy ovulatory	6 cycles	Windowed Herzog thresholds	47,394	16,299	34.4%	5.2%
healthy ovulatory	6 cycles	Windowed Herzog with minimum data	40,861	12,925	31.6%	18.3%
healthy ovulatory	12 cycles	Cycle reproducibility rule	15,706	355	2.3%	68.6%
healthy ovulatory	12 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	12 cycles	Negative-binomial regression, stabilized dispersion	45,088	2,155	4.8%	9.8%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	12 cycles	Windowed Herzog thresholds	48,682	12,364	25.4%	2.6%
healthy ovulatory	12 cycles	Windowed Herzog with minimum data	45,088	10,544	23.4%	9.8%
healthy ovulatory	36-month full diary	Cycle reproducibility rule	6,132	1	0.0%	87.7%
healthy ovulatory	36-month full diary	Exact Herzog 2004, any pattern	0	0	NA	100.0%
healthy ovulatory	36-month full diary	Negative-binomial regression, stabilized dispersion	48,359	1,966	4.1%	3.3%
healthy ovulatory	36-month full diary	Windowed Herzog thresholds	49,605	5,576	11.2%	0.8%
healthy ovulatory	36-month full diary	Windowed Herzog with minimum data	48,359	4,940	10.2%	3.3%
heterogeneous menstruating-age	1 month	Cycle reproducibility rule	0	0	NA	100.0%
heterogeneous menstruating-age	1 month	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	1 month	Negative-binomial regression, stabilized dispersion	0	0	NA	100.0%
heterogeneous menstruating-age	1 month	Windowed Herzog thresholds	38,083	19,748	51.9%	23.8%
heterogeneous menstruating-age	1 month	Windowed Herzog with minimum data	0	0	NA	100.0%
heterogeneous menstruating-age	3 months	Cycle reproducibility rule	0	0	NA	100.0%
heterogeneous menstruating-age	3 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	3 months	Negative-binomial regression, stabilized dispersion	0	0	NA	100.0%
heterogeneous menstruating-age	3 months	Windowed Herzog thresholds	45,297	23,184	51.2%	9.4%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
heterogeneous menstruating-age	3 months	Windowed Herzog with minimum data	0	0	NA	100.0%
heterogeneous menstruating-age	4 months	Cycle reproducibility rule	0	0	NA	100.0%
heterogeneous menstruating-age	4 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	4 months	Negative-binomial regression, stabilized dispersion	37,529	1,180	3.1%	24.9%
heterogeneous menstruating-age	4 months	Windowed Herzog thresholds	46,369	23,060	49.7%	7.3%
heterogeneous menstruating-age	4 months	Windowed Herzog with minimum data	37,530	17,825	47.5%	24.9%
heterogeneous menstruating-age	6 months	Cycle reproducibility rule	5,189	409	7.9%	89.6%
heterogeneous menstruating-age	6 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	6 months	Negative-binomial regression, stabilized dispersion	41,240	1,862	4.5%	17.5%
heterogeneous menstruating-age	6 months	Windowed Herzog thresholds	47,490	22,546	47.5%	5.0%
heterogeneous menstruating-age	6 months	Windowed Herzog with minimum data	41,240	18,769	45.5%	17.5%
heterogeneous menstruating-age	12 months	Cycle reproducibility rule	20,438	1,279	6.3%	59.1%
heterogeneous menstruating-age	12 months	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	12 months	Negative-binomial regression, stabilized dispersion	45,303	2,122	4.7%	9.4%
heterogeneous menstruating-age	12 months	Windowed Herzog thresholds	48,710	20,543	42.2%	2.6%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
heterogeneous menstruating-age	12 months	Windowed Herzog with minimum data	45,303	18,437	40.7%	9.4%
heterogeneous menstruating-age	3 cycles	Cycle reproducibility rule	0	0	NA	100.0%
heterogeneous menstruating-age	3 cycles	Exact Herzog 2004, any pattern	16,673	8,602	51.6%	66.7%
heterogeneous menstruating-age	3 cycles	Negative-binomial regression, stabilized dispersion	1,947	50	2.6%	96.1%
heterogeneous menstruating-age	3 cycles	Windowed Herzog thresholds	45,349	22,907	50.5%	9.3%
heterogeneous menstruating-age	3 cycles	Windowed Herzog with minimum data	1,947	1,140	58.6%	96.1%
heterogeneous menstruating-age	6 cycles	Cycle reproducibility rule	25,842	2,006	7.8%	48.3%
heterogeneous menstruating-age	6 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	6 cycles	Negative-binomial regression, stabilized dispersion	41,326	1,910	4.6%	17.3%
heterogeneous menstruating-age	6 cycles	Windowed Herzog thresholds	47,488	22,375	47.1%	5.0%
heterogeneous menstruating-age	6 cycles	Windowed Herzog with minimum data	41,326	18,672	45.2%	17.3%
heterogeneous menstruating-age	12 cycles	Cycle reproducibility rule	20,074	1,854	9.2%	59.9%
heterogeneous menstruating-age	12 cycles	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	12 cycles	Negative-binomial regression, stabilized dispersion	45,332	2,153	4.7%	9.3%
heterogeneous menstruating-age	12 cycles	Windowed Herzog thresholds	48,706	20,061	41.2%	2.6%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
heterogeneous menstruating-age	12 cycles	Windowed Herzog with minimum data	45,332	18,002	39.7%	9.3%
heterogeneous menstruating-age	36-month full diary	Cycle reproducibility rule	13,364	1,605	12.0%	73.3%
heterogeneous menstruating-age	36-month full diary	Exact Herzog 2004, any pattern	0	0	NA	100.0%
heterogeneous menstruating-age	36-month full diary	Negative-binomial regression, stabilized dispersion	48,333	2,032	4.2%	3.3%
heterogeneous menstruating-age	36-month full diary	Windowed Herzog thresholds	49,587	17,983	36.3%	0.8%
heterogeneous menstruating-age	36-month full diary	Windowed Herzog with minimum data	48,333	17,206	35.6%	3.3%

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74 Rows with fewer than 1,000 classifiable windows are retained for transparency but their positive counts and rates are suppressed to
 75 avoid over-interpretation.

76 Supplementary Table 4. C1/C2/C3 decomposition and C3-exclusion sensitivity.

Cohort	Definition	Classifiable windows	Positive windows	Apparent classification rate	Indeterminate
healthy ovulatory	Minimum-data rule, excluding C3	48,359	4,940	10.2% (9.9%-10.5%)	3.3%
healthy ovulatory	Windowed Herzog C1 only	49,451	3,893	7.9% (7.6%-8.1%)	1.1%
healthy ovulatory	Windowed Herzog C2 only	49,495	2,753	5.6% (5.4%-5.8%)	1.0%
healthy ovulatory	Windowed Herzog C3 only	0	0	NA (NA-NA)	100.0%
healthy ovulatory	Windowed Herzog thresholds	49,605	5,576	11.2% (11.0%-11.5%)	0.8%
healthy ovulatory	Windowed Herzog thresholds, excluding C3	49,605	5,576	11.2% (11.0%-11.5%)	0.8%

Cohort	Definition	Classifiable windows	Positive windows	Apparent classification rate	Indeterminate
healthy ovulatory	Windowed Herzog with minimum data	48,359	4,940	10.2% (9.9%-10.5%)	3.3%
heterogeneous menstruating-age	Minimum-data rule, excluding C3	48,333	5,183	10.7% (10.5%-11.0%)	3.3%
heterogeneous menstruating-age	Windowed Herzog C1 only	49,392	4,171	8.4% (8.2%-8.7%)	1.2%
heterogeneous menstruating-age	Windowed Herzog C2 only	49,482	2,788	5.6% (5.4%-5.8%)	1.0%
heterogeneous menstruating-age	Windowed Herzog C3 only	38,989	14,425	37.0% (36.5%-37.5%)	22.0%
heterogeneous menstruating-age	Windowed Herzog thresholds	49,587	17,983	36.3% (35.8%-36.7%)	0.8%
heterogeneous menstruating-age	Windowed Herzog thresholds, excluding C3	49,587	5,837	11.8% (11.5%-12.1%)	0.8%
heterogeneous menstruating-age	Windowed Herzog with minimum data	48,333	17,206	35.6% (35.2%-36.0%)	3.3%

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78 Supplementary Table 5. Negative-binomial stabilized versus window-only dispersion sensitivity.

Cohort	Observation window	Regression comparator	Classifiable windows	Apparent classification rate	Indeterminate
healthy ovulatory	36-month full diary	Negative-binomial regression, stabilized dispersion	48,359	4.1% (3.9%-4.2%)	3.3%
healthy ovulatory	36-month full diary	Negative-binomial regression, window-only dispersion	48,359	4.1% (3.9%-4.2%)	3.3%
heterogeneous menstruating-age	36-month full diary	Negative-binomial regression, stabilized dispersion	48,333	4.2% (4.0%-4.4%)	3.3%
heterogeneous menstruating-age	36-month full diary	Negative-binomial regression, window-only dispersion	48,333	4.2% (4.0%-4.4%)	3.3%

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80 Supplementary Table 6. Study-level Monte Carlo results.

Cohort	Definition	Denominator	Mean apparent prevalence	2.5th-97.5th percentiles	P >=39.1%	P >=44.2%
healthy ovulatory	Cycle reproducibility rule	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
healthy ovulatory	Cycle reproducibility rule	Classifiable participants only	NA	NA-NA	NA	NA
healthy ovulatory	Negative-binomial regression, stabilized dispersion	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
healthy ovulatory	Negative-binomial regression, stabilized dispersion	Classifiable participants only	NA	NA-NA	NA	NA
healthy ovulatory	Windowed Herzog thresholds	All participants	37.5%	20.0%-53.3%	45.9%	19.7%
healthy ovulatory	Windowed Herzog thresholds	Classifiable participants only	41.4%	23.1%-60.0%	60.8%	39.5%
healthy ovulatory	Windowed Herzog with minimum data	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
healthy ovulatory	Windowed Herzog with minimum data	Classifiable participants only	NA	NA-NA	NA	NA
heterogeneous menstruating-age	Cycle reproducibility rule	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
heterogeneous menstruating-age	Cycle reproducibility rule	Classifiable participants only	NA	NA-NA	NA	NA
heterogeneous menstruating-age	Negative-binomial regression, stabilized dispersion	All participants	0.0%	0.0%-0.0%	0.0%	0.0%
heterogeneous menstruating-age	Negative-binomial regression, stabilized dispersion	Classifiable participants only	NA	NA-NA	NA	NA
heterogeneous menstruating-age	Windowed Herzog thresholds	All participants	46.0%	30.0%-63.3%	79.9%	54.5%
heterogeneous menstruating-age	Windowed Herzog thresholds	Classifiable participants only	50.8%	32.0%-69.2%	89.4%	75.9%
heterogeneous menstruating-age	Windowed Herzog with minimum data	All participants	0.0%	0.0%-0.0%	0.0%	0.0%

Cohort	Definition	Denominator	Mean apparent prevalence	2.5th-97.5th percentiles	P >=39.1%	P >=44.2%
heterogeneous menstruating-age	Windowed Herzog with minimum data	Classifiable participants only	NA	NA-NA	NA	NA

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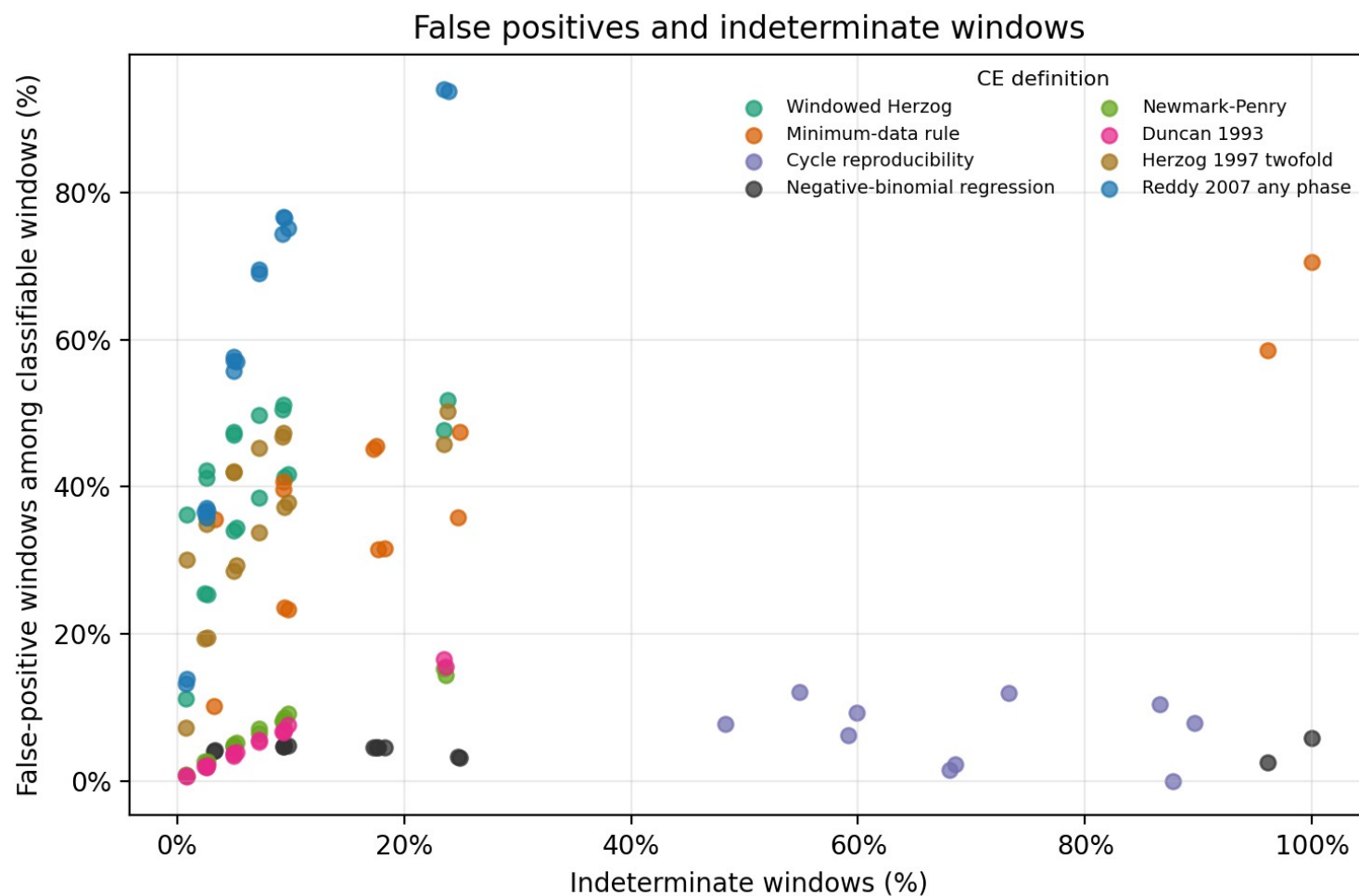
82 Supplementary Table 7. Assumption-based historical definitions.

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
healthy ovulatory	3 months	Duncan 1993 ten-day rule	45,280	3,163	7.0% (6.8%-7.2%)	9.4%
healthy ovulatory	3 months	Herzog 1997 twofold rule	45,280	16,862	37.2% (36.8%-37.7%)	9.4%
healthy ovulatory	3 months	Newmark-Penry perimenstrual rule	45,280	3,939	8.7% (8.4%-9.0%)	9.4%
healthy ovulatory	3 months	Newmark-Penry two-thirds sensitivity	45,280	1,870	4.1% (4.0%-4.3%)	9.4%
healthy ovulatory	3 months	Reddy 2007 any-phase twofold rule	45,280	34,681	76.6% (76.2%-77.0%)	9.4%
healthy ovulatory	36-month full diary	Duncan 1993 ten-day rule	49,605	361	0.7% (0.7%-0.8%)	0.8%
healthy ovulatory	36-month full diary	Herzog 1997 twofold rule	49,605	3,603	7.3% (7.0%-7.5%)	0.8%
healthy ovulatory	36-month full diary	Newmark-Penry perimenstrual rule	49,605	427	0.9% (0.8%-0.9%)	0.8%
healthy ovulatory	36-month full diary	Newmark-Penry two-thirds sensitivity	49,605	170	0.3% (0.3%-0.4%)	0.8%
healthy ovulatory	36-month full diary	Reddy 2007 any-phase twofold rule	49,605	6,599	13.3% (13.0%-13.6%)	0.8%
heterogeneous menstruating-age	3 months	Duncan 1993 ten-day rule	45,297	3,018	6.7% (6.4%-6.9%)	9.4%
heterogeneous menstruating-age	3 months	Herzog 1997 twofold rule	45,297	21,417	47.3% (46.8%-47.7%)	9.4%
heterogeneous menstruating-age	3 months	Newmark-Penry perimenstrual rule	45,297	3,696	8.2% (7.9%-8.4%)	9.4%
heterogeneous menstruating-age	3 months	Newmark-Penry two-thirds sensitivity	45,297	1,733	3.8% (3.7%-4.0%)	9.4%

Cohort	Observation window	Definition	Classifiable windows	False-positive windows	False-positive rate	Indeterminate
heterogeneous menstruating-age	3 months	Reddy 2007 any-phase twofold rule	45,297	34,750	76.7% (76.3%-77.1%)	9.4%
heterogeneous menstruating-age	36-month full diary	Duncan 1993 ten-day rule	49,587	311	0.6% (0.6%-0.7%)	0.8%
heterogeneous menstruating-age	36-month full diary	Herzog 1997 twofold rule	49,587	14,914	30.1% (29.7%-30.5%)	0.8%
heterogeneous menstruating-age	36-month full diary	Newmark-Penry perimenstrual rule	49,587	378	0.8% (0.7%-0.8%)	0.8%
heterogeneous menstruating-age	36-month full diary	Newmark-Penry two-thirds sensitivity	49,587	175	0.4% (0.3%-0.4%)	0.8%
heterogeneous menstruating-age	36-month full diary	Reddy 2007 any-phase twofold rule	49,587	6,854	13.8% (13.5%-14.1%)	0.8%

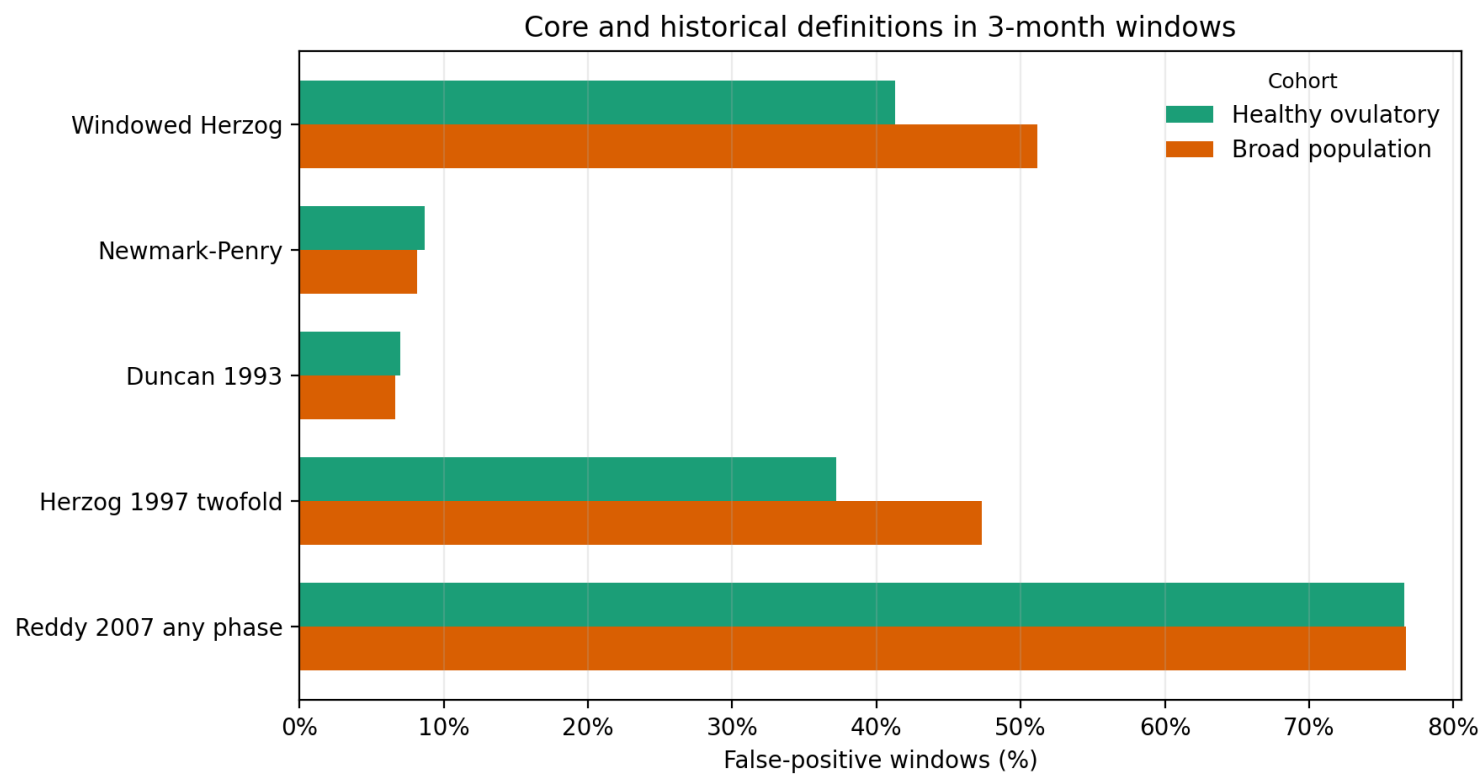
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84 Supplementary Figure 1. False-positive and indeterminate frontier.



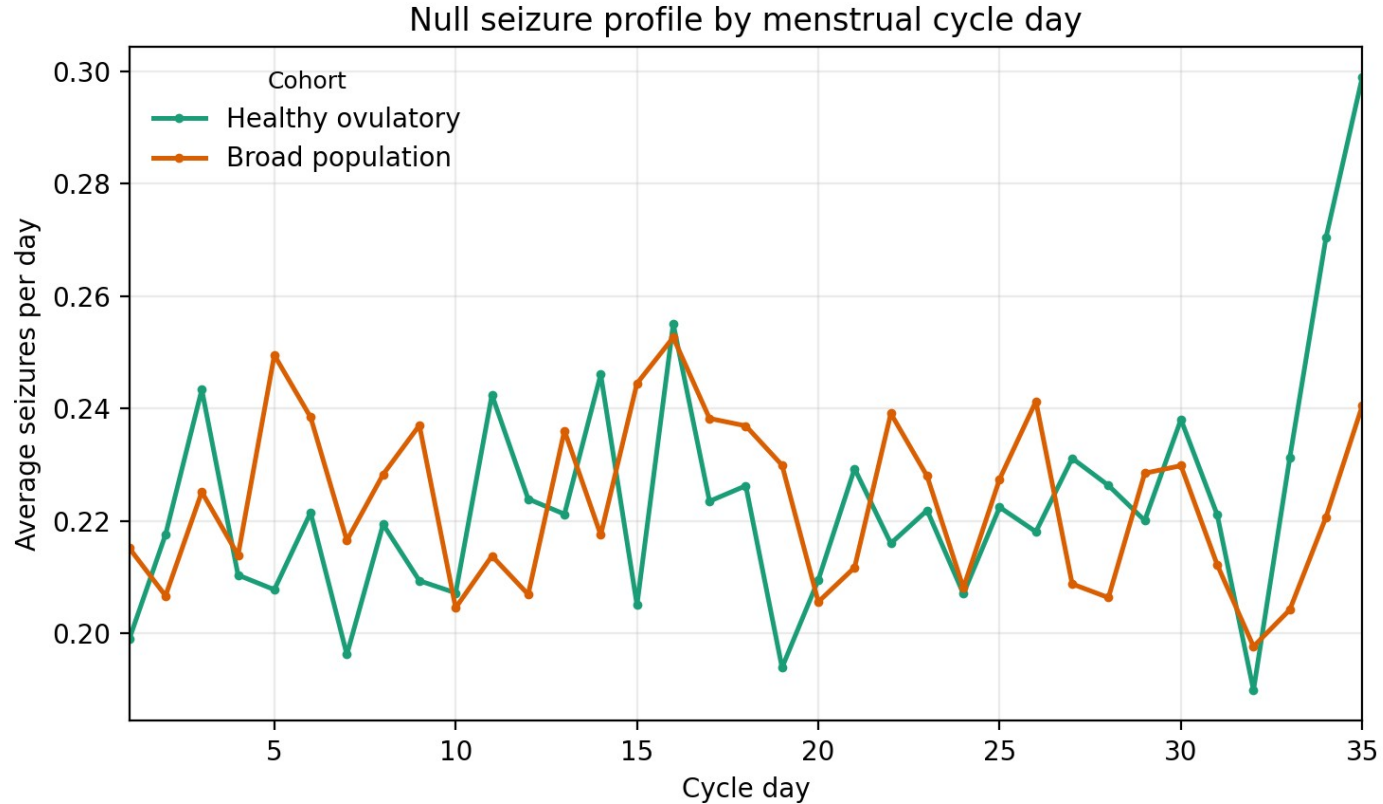
85

86 Supplementary Figure 2. Historical and core definitions in 3-month windows.



87

88 Supplementary Figure 3. Null seizure profile by menstrual cycle day.



89

90 A8. Output provenance and assumptions

- 91 • Definition D uses a participant-full-diary method-of-moments negative-binomial alpha recorded in `d_alpha`; Poisson robust
- 92 fallback is recorded in `d_reason` when statsmodels negative-binomial fitting fails. `D_window_alpha` re-estimates alpha from the
- 93 analyzed window as a non-oracle sensitivity.

- 94 • Healthy ovulatory cohort used hormone_cycler build_patient_profile/render_cycle with ovulation_probability set to 1.0 because
95 simulate_diary did not expose a public force-ovulation knob.
- 96 • Historical definitions H1-H4 are assumption-based operationalizations and are flagged in summary outputs.
- 97 • Study-level Monte Carlo samples each selected participant from a deterministic pool of precomputed random valid 3-month
98 windows to avoid retaining all daily diaries in memory.
- 99 • The hormone simulator exposes medical-factor controls but no natural prevalence sampler; heterogeneous-cohort medical factors
100 were sampled from config.yaml rates.
- 101 • Positive-control true-coupling modes add extra Poisson seizure events on target phase days after diary alignment and write to
102 separate output directories.